

A DIFFERENT VIEW OPEN ACCESS

Child Psychiatry and Child Neurology Need to Come Together

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An unfavourable development within Child Psychiatry and Child Neurology is that these two disciplines have been separated since the beginning of the twentieth century. Unfortunately, no serious attempt has been successful in re-joining these two related branches of medicine.

The Psychiatry-Neurology split was occasioned by the elaboration of a theory of dreams when, beginning in 1895, Sigmund Freud, training as a neurologist, came to the conclusion that dreams were purely psychological. However, in 1914 he reconsidered, declaring: 'we must recollect that all our provisional ideas in psychology will presumably one day be based on organic substructure' [1]. In that sense, he saw the eventual merging of the two disciplines. That time is long overdue.

1 | Neuropaediatric and Neuropsychiatric Disorders

Young patients with behavioural—or clear neurodevelopmental/neuropsychiatric symptoms—go to one or the other type of physician: a child psychiatrist, a paediatrician, or a neuropaediatrician. The result may be a delayed diagnosis and delayed or denied intervention. Many 'neurological' disorders present with psychiatric symptoms, including a number of genetic syndromes: Fragile X, Neurofibromatosis type 1, XXY-syndrome, XYY-syndrome and many microdeletion/microduplication syndromes. These all need neuropsychiatric as well as neurological assessment. On the other hand, children with classical neuropaediatric disorders and disabilities, including epilepsy, cerebral palsy [2] and neuromuscular disorders, very often have concomitant neuropsychiatric or neurodevelopmental conditions or

difficulties, particularly attention-deficit/hyperactivity disorder (ADHD), autism, intellectual disability, and borderline intellectual functioning. For patients and their families, there is a clear advantage if assessments and treatments are collaborative and coherent, so as to minimise referrals to other clinics, involving waiting times and meetings with new professionals.

2 | A Wide Range of Medical Comorbid Conditions in ADHD

A broad perspective regarding medical conditions and ADHD has been presented in an article based on a collaborative work from many parts of the world. In this comprehensive systematic reviews and meta-analyses of comorbid health conditions in people with ADHD, varied levels of evidence for an association between ADHD and multiple health conditions were found. The thought-provoking conclusion was that physicians should be aware of the wide range of neurological, psychiatric, health conditions associated with ADHD to ensure a comprehensive and holistic assessment of the needs of individuals with ADHD. Clinicians should consider a wide range of neurological, psychiatric, suicide-related, sleep, metabolic, musculoskeletal, oral, allergic and visual conditions, as well as the increased risk of mortality when assessing individuals with ADHD [3].

3 | ADHD and Obesity

Children and adolescents with obesity, with their basic intervention and follow-up at a paediatric clinic, also need a neurodevelopmental/neuropsychiatric assessment, with a view to identifying

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possibly co-occurring ADHD, sometimes in association with borderline intellectual function or mild intellectual disability. In a study by Cortese and Tessari [4], possible mechanisms underlying the link between ADHD and obesity were addressed, highlighting the role, among others, of abnormal eating patterns, sedentary lifestyle and possible common genetic alterations.

4 | ADHD and/or Autism—Follow-Up in Young Adulthood

In a follow-up study of 77 women, mean age of 27 years, diagnosed with ADHD and/or autism before age 18 years, chronic pain was reported by 59 (76.6%) of the women. An important result was that treatment for ADHD might reduce the pain in some cases. The findings highlighted the importance of health professionals considering possible unrecognised ASD and/or ADHD in women with chronic pain [5].

5 | Pain in Children

In a study based on 10–11-year-old Swedish children, weekly pain was reported among 53% of children with symptoms of ADHD, combined type, compared to 36% of children without symptoms of ADHD. Hyperactivity and impulsivity had a stronger association to pain than did inattention-related problems. The frequent occurrence and the association between pain and neurodevelopmental conditions among children was emphasised and that it could complicate both the clinical picture and the treatment [6].

6 | Paediatric Acute-Onset Neuropsychiatric Syndrome

Another clear example of the necessity of collaboration between child psychiatry/neuropsychiatry and paediatrics/neuropaediatrics, together with paediatric immunology and rheumatology, is the paediatric acute-onset neuropsychiatric syndrome (PANS) [7].

7 | Collaboration Between Child Neuropsychiatry and Neuropaediatrics

Child psychiatrists and neuropaediatricians within the area of child neuropsychiatry and neuropaediatrics should collaborate and work ‘side by side’. Autism, ADHD and intellectual disability, as well as many other neurodevelopmental conditions, need a medical work up with regard to aetiology [8]. All children and adolescents with neuropsychiatric/neurodevelopmental conditions, with or without—at the time of assessment—a verified aetiology, need clinical follow-up. Symptoms may change, increase or decrease and new symptoms may emerge—and, in addition, medical technology will develop and may make it possible to later identify an aetiology.

Clearly the divide between psychiatry and neurology needs to be closed [9]. What we have discussed here in terms of Child Psychiatry and Child Neurology also applies to the bridges between Adult Psychiatry and Adult Neurology services. There has been an unfortunate division of the brain between the fields

of these medical specialities. From a healthcare perspective, and from a child and adolescent perspective, this means that we need to bridge the gap between Child and Adolescent Mental Health/Child Psychiatry—and Paediatrics/Neuropaediatrics so as to connect these units with each other—with the aim of benefitting the development and health of affected children [10].

Author Contributions

Mary Coleman: methodology, data curation, writing – review and editing, conceptualization, writing – original draft. **Elisabeth Fernell:** conceptualization, methodology, data curation, writing – original draft, writing – review and editing. **Maria Råstam:** conceptualization, methodology, data curation, writing – original draft, writing – review and editing. **Helen Minnis:** conceptualization, data curation, writing – original draft, writing – review and editing, methodology. **Christopher Gillberg:** conceptualization, methodology, data curation, writing – original draft, writing – review and editing.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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